

Soft supermolecule stabilized buried interface for high-performance inverted perovskite solar cells and modules

Received: 12 December 2025

Accepted: 27 May 2026

Published online: 11 July 2026

Cite this article as: Yang, R., Ding, J., Hou, T. *et al.* Soft supermolecule stabilized buried interface for high-performance inverted perovskite solar cells and modules. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-74018-8>

Ruoqi Yang, Jike Ding, Tian Hou, Yue Yu, Ziyuan Liu, Hao Liu, Lang Yu, Chongzhu Hu, Shaokuan Gong, Xihan Chen, Xilai He, Kun Zhang, Meirong Fu, Yang Wang, Xinxing Liu, Dongmei He, Xuxia Shai, Jiajia Zhang, Xingyu Gao, Xuanhua Li, Yuelong Huang, Cong Chen, Jianhong Yi & Jiangzhao Chen

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Soft supermolecule stabilized buried interface for high-performance inverted perovskite solar cells and modules

Ruoqi Yang^{1#}, Jike Ding^{2#}, Tian Hou^{3#}, Yue Yu^{1*}, Ziyuan Liu¹, Hao Liu², Lang Yu³, Chongzhu Hu³, Shaokuan Gong⁴, Xihan Chen^{4*}, Xilai He⁵, Kun Zhang^{6,7,8}, Meirong Fu⁹, Yang Wang^{6,7,8*}, Xinxing Liu¹, Dongmei He¹, Xuxia Shai¹, Jiajia Zhang¹⁰, Xingyu Gao⁹, Xuanhua Li^{5*}, Yuelong Huang^{3*}, Cong Chen^{2*}, Jianhong Yi¹ & Jiangzhao Chen^{1*}

¹Faculty of Materials Science and Engineering, Kunming University of Science and Technology, Kunming 650093, China

²State Key Laboratory of Reliability and Intelligence of Electrical Equipment, School of Materials Science and Engineering, Hebei University of Technology, Tianjin 300401, China

³Huzhou Phoenixolar Co.,Ltd., Huzhou 313200, China

⁴State Key Laboratory of Quantum Functional Materials, SUSTech Energy Institute for Carbon Neutrality, Department of Mechanical and Energy Engineering, Southern University of Science and Technology, Shenzhen 518055, China

⁵State Key Laboratory of Solidification Processing, Center for Nano Energy Materials, School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an 710072, China

⁶Key Laboratory of Green Printing, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China.

⁷CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China.

⁸University of Chinese Academy of Sciences, Beijing 100049, China.

⁹Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201204, China

¹⁰Anhui Provincial Key Laboratory of Green Carbon Chemistry, College of Chemistry and Materials Engineering, Fuyang Normal University, Fuyang 236037, China

[#]These authors contributed equally to this work.

Corresponding authors

E-mail: YueYu@kust.edu.cn (Yue Yu); chenxh@sustech.edu.cn (Xihan Chen); wangyang@iccas.ac.cn (Yang Wang); lixh32@nwpu.edu.cn (Xuanhua Li); hyl@phoenixolar.com (Yuelong Huang); chencong@hebut.edu.cn (Cong Chen); jzchen@kust.edu.cn (Jiangzhao Chen)

Abstract

Self-assembled molecules (SAMs) have emerged as a promising hole transport material for improving the power conversion efficiency (PCE) of p-i-n inverted perovskite solar cells (PSCs). However, molecular

aggregation and insufficient coverage of SAMs, the defects at the bottom surface of perovskite films, and weak adhesive force of perovskite films on SAMs result in poor quality and longevity of the interface between SAMs and perovskite (buried interface), hampering the realization of long-term operationally stable inverted PSCs. Here we report a supramolecular host-guest interaction strategy to stabilize buried interface in inverted PSCs. Through incorporating sulfonyl-functionalized calixarene molecules, namely 4-tert-Butylsulfonylcalix[4]arene (SC4A) with soft Lewis basicity, the reinforced buried interface durability is fulfilled by homogenizing SAM film, passivating interface defects, releasing interface residual stress, and bilateral chemical bridging. The resulting inverted PSCs and large-area modules accomplish a certified PCE of 27.12% and 22.25% (aperture area 655.2 cm²), respectively. Moreover, the SC4A-modulated devices retain 97.4% of its initial PCE after continuous operation under one sun illumination for 2125 h, and 90.7% of its original PCE after damp heat aging (85 °C and 85% relative humidity) for 2000 h.

Keywords: perovskite solar cells; buried interface; self-assembled molecules; soft supramolecule; stability

Introduction

Inverted (p-i-n) perovskite solar cells (PSCs) have become a forefront focus due to their competitive power conversion efficiencies (PCEs), easy fabrication, high stability, and good compatibility with various tandem solar cells¹⁻⁷. The dramatic development of self-assembled molecules (SAMs) has significantly improved the PCEs of inverted PSCs, owing to efficient hole extraction, interfacial defect passivation capability, and energy level modulation⁸⁻¹⁶. However, commonly used and sophisticated SAMs, such as [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), and [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) suffer from molecular aggregation, inhomogeneous distribution and insufficient coverage on the substrates of nickel oxide or transparent conductive electrodes¹⁷⁻¹⁹. In addition, presently most of state-of-the-art SAMs cannot effectively passivate the defects at the bottom surface of perovskite films because of lacking appropriate functional groups at the end of SAMs which can chemically interact with perovskites^{10,14,17,20}. The uncontrollable

perovskite crystallization and inhomogeneous SAMs usually result in a large number of defects, residual stress, nanovoids and non-uniformity at the interface between hole transport layer (HTL) and perovskite films (called buried interface)^{17,21,22}. Finally, the perovskite films are easy to divorce from HTL due to weak adhesive force resulting from absent strong chemical bonding between the two functional layers. These disadvantages ultimately would lead to poor quality and stability of buried interface, thereby aggravating interfacial nonradiative recombination and deteriorating the device photovoltaic performance and long-term operational stability, especially when manufacturing large-area devices²³⁻²⁵.

To overcome the above challenges, various strategies have been developed, principally including design and synthesis of SAMs^{8,14-16,26,27}, modulation of commonly employed SAMs¹⁷⁻¹⁹, and manipulation of interface between SAM and perovskite layer^{21,22}. Among them, from the perspective of function-driven molecular design, meticulous modulation of buried interface is a promising approach to enhance coverage rate and uniformity of SAMs, passivate interfacial defects, realize interfacial chemical bridging and ameliorating buried interface quality and durability^{21,22}. To simultaneously achieve the above multiple functions, it is highly desirable to leverage chemical bridging molecules with multiple active sites and high dipole moment, which synchronously strongly interact with SAMs and perovskites, to regulate buried interface²⁸⁻³⁰. Specifically, supramolecules, such as crown ethers^{31,32}, cyclodextrins³³, calixpyrroles³⁴, and calixarenes^{35,36}, are one kind of multisite organic functional molecules, providing a versatile toolbox to precisely manipulate the functional layers and interface in PSCs via multifarious chemical bonds primarily involving hydrogen bonding, coordination bonding, π - π interactions, van der Waals forces, and host-guest chemistry³⁷. It has been demonstrated that supramolecule modulators can regulate perovskite crystallization³⁸, passivate defects³⁹⁻⁴², suppress ion migration^{34,35,43}, manage residual PbI_2 ⁴⁴, releasing residual stress⁴⁵, inhibiting lead leakage³³ and suppressing halide segregation³², heightening PCE and longevity of PSCs. It was reported that based on heteroatom functionalization strategy and cavity size effect, the modulation effect of supramolecules can be controlled by tuning their Lewis basicity and thus interaction strength with perovskites, thereby improving device performance^{35,41,46-48}. To date, supramolecules are mostly used to modulate perovskite/electron transport layer interface in inverted PSCs^{35,49}, perovskite/charge transport layer interface in n-i-p PSCs^{31,38,45,50}, or are directly introduced into perovskite precursor solution^{32,33,39,41,51}. To the best of our knowledge,

however, the study on modulation of buried interface with supramolecules in inverted PSCs has not been reported yet. Therefore, there is an urgent need to tackle the above-mentioned intricate challenges at buried interface through host-guest interaction from the viewpoint of supramolecular engineering.

Here we reported a supramolecular host-guest interaction strategy to stabilize buried interface of inverted PSCs with minimized interfacial energy losses. A series of calixarene supramolecules with or without heteroatoms and groups containing heteroatoms, namely 4-tert-Butylcalix[4]arene (C4A), 4-tert-Butylthiacalix[4]arene (TC4A), and 4-tert-Butylsulfonylcalix[4]arene (SC4A), were utilized to manipulate the buried interface. Considering both Lewis basicity and dipole moment, SC4A with sulfonyl functionalization exhibited the optimal interface modulation effect according to hard-soft acid-base (HSAB) theory. Incorporating SC4A can suppress SAMs aggregation, promote homogeneous distribution of SAMs, passivate interface defects, release interface residual stress, reduce interface nanovoids, as well as improve the quality and stability of buried interface, by simultaneous chemical interaction with both perovskites and SAMs via hydrogen bonding, coordination bonding and van der Waals forces. As a result, the SC4A-modulated PSCs and large-area modules accomplished a PCE of 27.36% (certified steady-state efficiency of 27.12%) and certified 22.25%, respectively, which are among the most advanced inverted PSCs and modules. After SC4A modification, the reinforced buried interface stability was translated into saliently enhanced device stability, maintaining 97.4% of its initial PCE after continuous maximum power point tracking (MPPT) under 1-sun illumination for 2125 h, and 90.7% of its original PCE after aging under damp heat aging conditions (85 °C and 85% relative humidity) for 2000 h.

Results

Theoretical screening of calixarene supramolecules

To unleash the full potential of calixarenes in PSCs, we exploited sulfur atom (S) and sulphonyl to functionalize the C4A reference molecules to TC4A and SC4A, respectively, aiming to increase the Lewis base softness and dipole moment, which amplifies its interaction with perovskites and SAMs^{41,46,47}. First, density functional theory (DFT) was performed to calculate the electrostatic potential (ESP) and dipole moments of three supramolecules. The negatively charges of C4A is mainly distributed around the oxygen

atoms at the bottom of the cup (Fig. 1a). After the functionalization of S atoms, the negative potential of TC4A is significantly transferred to the sulfur atoms. Notably, the introduction of the strong electron-withdrawing sulphonyl groups in SC4A causes the electron cloud density to be highly concentrated around -OH and sulphonyl groups, forming a highly negatively charged region. This indicates that the introduction of -SO₂- disrupts the original electron delocalization system of calixarene, thereby inducing redistribution of the electron cloud density in the vicinity of the hydroxyl and sulfonyl groups. This electronic reorganization further modulates the polarity characteristics of the molecule. SC4A exhibited a higher dipole moment of 16.77 D compared to those of C4A (3.38 D) and TC4A (7.80 D). The increased molecular polarity may ameliorate the energy level alignment at HTL/perovskite interface and accordingly promote hole extraction⁵².

In our work, mixed SAMs of Me-4PACz and MeO-2PACz as well as [4-(7H-dibenzo[c,g]carbazol-7-yl)butyl]phosphonic acid (4PADCB) with different dipole moments were adopted (Supplementary Fig. 1). Thus, we investigated the interaction of SAMs with perovskites. The DFT results revealed that the binding energy between SAMs and the perovskites was relatively small, resulting in a very weak adhesive force of perovskite films on the surface of HTL, which is not conducive to long-term stability of buried interface and devices (Supplementary Fig. 2). Subsequently, we explored the interactions of three calixarenes with perovskites and SAMs. The undercoordinated Pb²⁺ is the primary defects at the buried interface and shows a soft Lewis acid nature. In the chalcogen group, the softness of the base increases with the increase of the atomic number⁵²⁻⁵⁴. Owing to the high electronegativity and substantial electron cloud density of the sulfonyl oxygen atom, it was inferred that SC4A showed more pronounced soft base characteristics than C4A and TC4A. According to HSAB theory^{52,53}, the SC4A can form more stronger coordination bonds with Pb²⁺ compared with C4A and TC4A. As shown in Fig. 1b, this speculation was also evidenced by gradually increased binding energy (E_b) in the order of C4A (-1.40 eV), TC4A (-1.64 eV) and SC4A (-1.70 eV). The X-ray photoelectron spectroscopy (XPS) results further affirmed the strongest bonding of SC4A as evidenced by the largest Pb 4*f* shift among all calixarenes (Supplementary Fig. 3). Specifically, it is evident from the charge density difference map that the calixarene was arranged in an "upside-down" mode. That is, the four -OH groups were toward perovskite, coordinating with Pb²⁺, while the four tert-butyl groups were toward SAMs, interacting with carbazole in SAMs via van der Waals

forces, thereby chemically bridging SAMs layer and perovskite layer (Fig. 1b-d). The binding energy of C4A, TC4A and SC4A with Me-4PACz was -0.58 eV, -0.66 eV and -0.67 eV, respectively, suggesting the strongest interaction of SC4A with Me-4PACz (Fig. 1c). The binding energy of SC4A with MeO-2PACz was -0.69 eV, larger than those of C4A (-0.58 eV) and TC4A (-0.66 eV), which is indicative of the strongest binding of SC4A with MeO-2PACz (Fig. 1d). Similarly, the binding energy between SC4A and 4PADCB was also the strongest (Supplementary Fig.4). The above results show that SC4A exhibited the strongest interaction with both perovskite and SAMs, which can fulfil chemical bridge between perovskite layer and SAMs layer, and thus strengthen the durability of buried interface. Meanwhile, SC4A could also heal the defects (e.g., undercoordinated Pb^{2+} and iodide vacancies) at the surface of perovskite films, suppressing non-radiative recombination and improving buried interface quality. In addition, SC4A should be able to inhibit the aggregation of SAMs by van der Waals forces and/or hydrogen bonding between them.

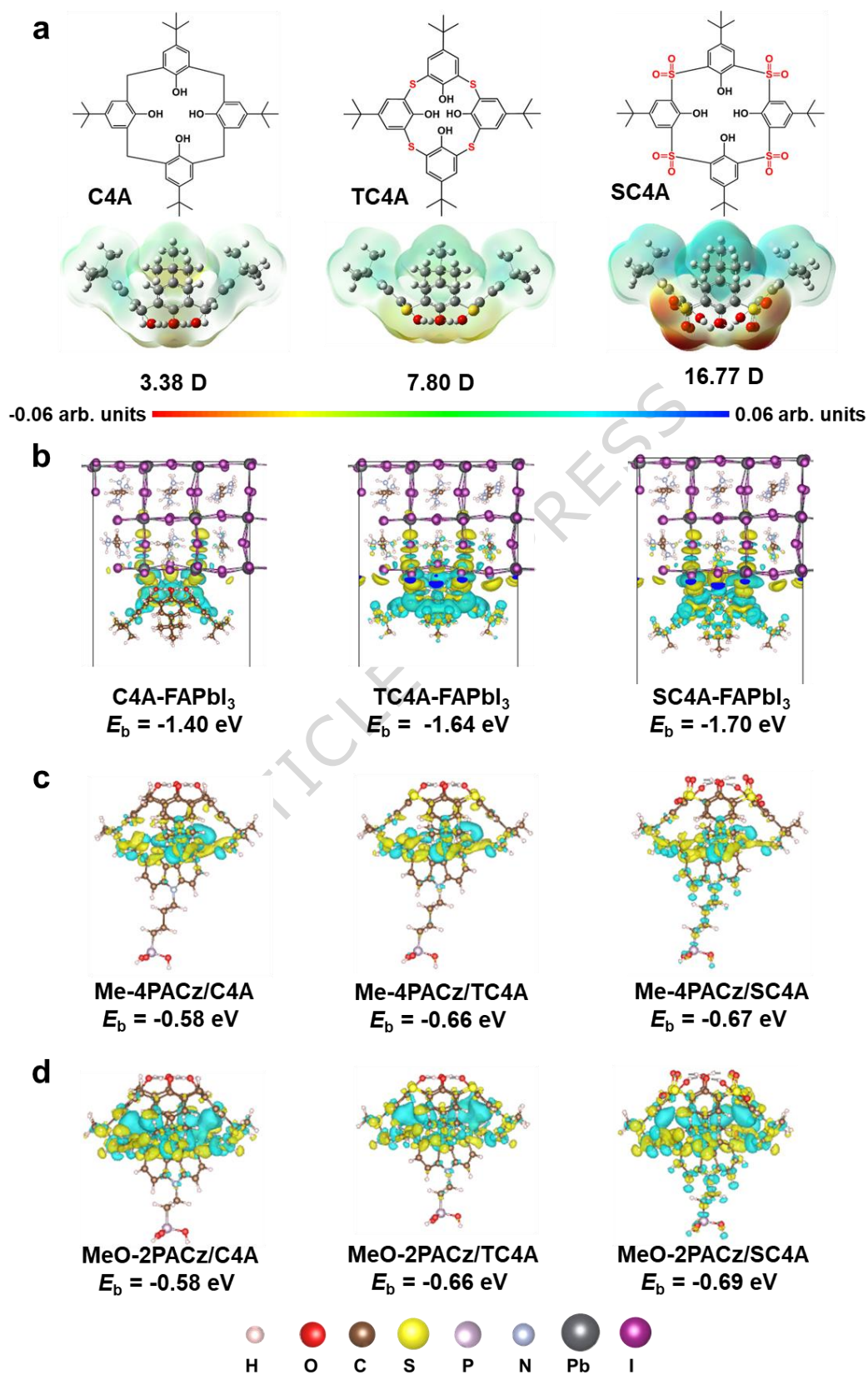


Fig. 1| Theoretical calculations on the interaction of three supramolecules with SAM molecules and perovskite. a, Chemical structures and ESP distribution of C4A, TC4A and BC4A. b, Binding energy of

C4A, TC4A, and SC4A molecules with perovskite. **c,d**, Binding energy of C4A, TC4A, and SC4A molecules absorbed onto Me-4PACz (**c**) and MeO-2PACz (**d**).

Interaction between SC4A and SAM

Fourier transform infrared spectroscopy (FTIR) was carried out to explore the interaction between SC4A and SAMs. The stretching vibration peak at 3411 cm^{-1} of -OH in SC4A was shifted to 3488 cm^{-1} after mixing with SAMs (Fig. 2a). Meantime, the SAM exhibited two peaks at 1111 cm^{-1} and 1067 cm^{-1} assigned to the stretching vibration peaks of P=O and P-OH, respectively, which were shifted to 1134 cm^{-1} and 1086 cm^{-1} after mixing with SC4A, respectively (Fig. 2b). It implies that the hydrogen bonding was formed between -OH groups in SC4A and phosphonic acid groups in SAMs. The binding mode of the SC4A molecule with 4PADCB is similar to that of Me-4PACz and MeO-2PACz (Supplementary Fig. 5), indicating that SC4A can serve as a universal interface regulation platform for carbazole-based phosphoric acid SAMs, providing a widely applicable solution for constructing high-performance inverted PSCs. In addition, from XPS in Fig. 2c and d, the shift of binding energies of P 2p peaks in SAMs and S 2p in SC4A further confirmed the hydrogen bonding between the hydroxyl groups in the SC4A and phosphonic acid groups in SAMs. To gain deep insights into the intermolecular interactions between SC4A and SAM, we performed nuclear magnetic resonance spectroscopy (NMR) analysis on different molar ratios of Me-4PACz and SC4A (Fig. 2e). Upon the gradual increase of Me-4PACz content in SC4A solution, the proton peak on -OH in SC4A shifted toward low-field direction. Combined with the ^{31}P NMR measurements (Supplementary Fig. 6), it was observed that the ^{31}P peak shifted upward after the addition of SC4A, which once again corroborated the hydrogen bonding interaction between -OH in SC4A and phosphonic acid group in Me-4PACz. Upon increasing Me-4PACz ratio in SC4A solution, the proton peak on benzene ring shifted toward low field, while upon increasing SC4A content in Me-4PACz solution, the proton peaks on carbazole and the -N-CH₂- exhibited a high-field shift, which could be attributed to π - π interactions and/or van der Waals forces interactions of tertiary butyl or benzene ring in SC4A with carbazole in Me-4PACz, consistent with theoretical calculation results. Furthermore, the same interactions also appeared between SC4A and MeO-2PACz (Supplementary Fig. 7). In short, the interaction of SC4A with SAMs should be beneficial for restraining SAM aggregation and facilitating homogeneous distribution and ordered arrangement of SAMs on NiO_x films.

Effect of SC4A modification on SAM films

We further examined the impact of SC4A treatment on the uniformity and compactness of SAMs films. As revealed by atomic force microscopy (AFM) images (Supplementary Fig. 8), the smaller root-mean square (RMS) of SAM/SC4A film than pristine SAM film confirmed improved flatness and denseness, which is conducive to good interface contact and uniform coverage of the perovskite film. This uniformity has also been confirmed in the large-area coated films (Supplementary Figs. 9, 10 and Supplementary Table 10). The surface contact potential difference (CPD) of the SAM and SAM/SC4A film (Fig. 2f, g, Supplementary Fig. 11) were conducted. As presented in Fig. 2f, g and Supplementary Fig. 11, Kelvin probe force microscopy (KPFM) measurements further revealed that the SAM/SC4A film displayed a more uniform and smaller surface potential of -338 mV compared to the SAM film (-298 mV), indicating a more p-type surfaces after SC4A modification. After SC4A modulation, the bandgaps (E_g) of perovskite films (1.53 eV) and HTLs (3.86 eV) were invariant (Supplementary Fig. 12). Then, ultraviolet photoelectron spectroscopy (UPS) measurements were further conducted to accurately calculate the energy levels of perovskite films and HTLs (Supplementary Fig. 13). It can be seen from Fig. 2h that the valence band maximum (E_{VBM}) of the HTL without or with SC4A was -5.36 eV and -5.42 eV, respectively. The control and SC4A-modified perovskite films possessed a E_{VBM} of -6.08 eV and -5.94 eV, respectively. The E_{VBM} offset between SC4A modified HTL and SC4A modified perovskite film was 0.52 eV, which is larger than that between pristine HTL and pristine perovskite film (0.72 eV), suggesting reduced interfacial energy barrier and thus accelerated hole extraction.

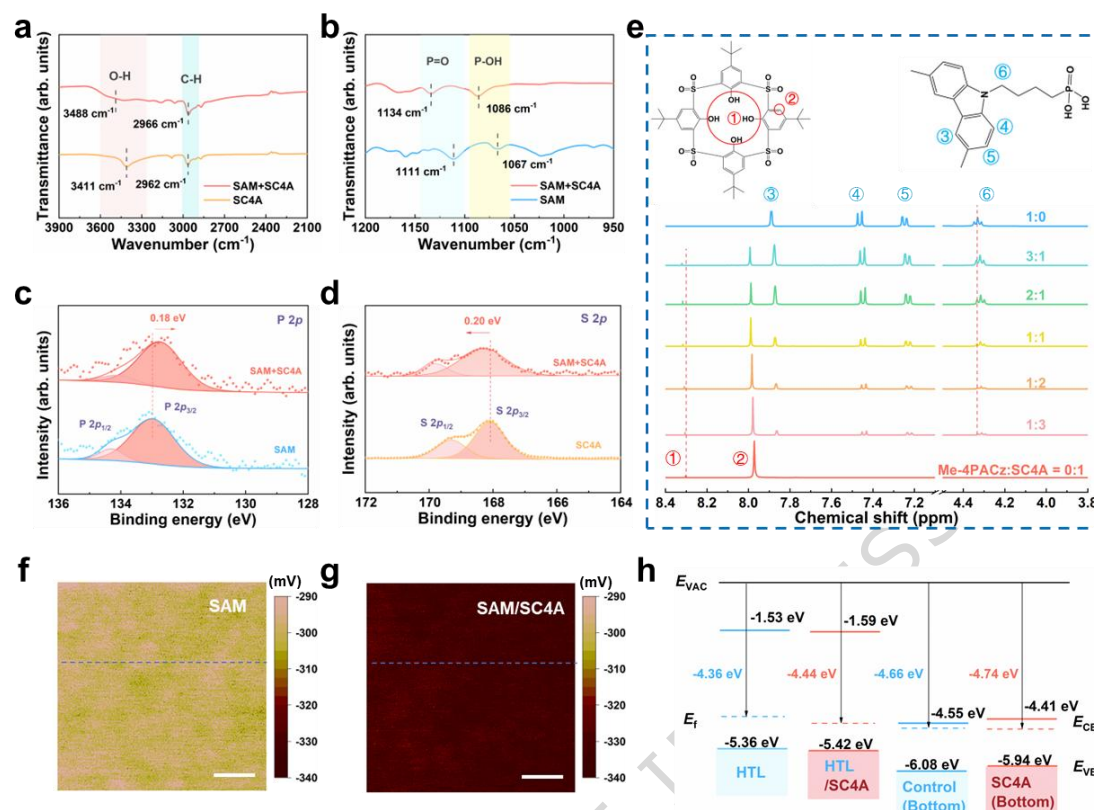


Fig. 2 | Effect of SC4A treatment on SAM films. **a,b**, FTIR spectra of SAM, SC4A and SC4A+SAM. **c**, XPS spectra of P 2*p* core levels of SAM and SAM+SC4A. **d**, XPS spectra of S 2*p* core levels of SC4A and SAM+SC4A. **e**, ¹H NMR spectra of mixtures of SC4A and Me-4PACz with different molar ratios. KPFM of NiO_x/SAM (**f**) and NiO_x/SAM/SC4A (**g**), Scale bars are 200 nm. **h**, The energy band arrangement diagram of HTLs and perovskite films with or without SC4A.

Interaction of SC4A with perovskite

Except for the interaction of SC4A with SAMs, we also studied the interaction between SC4A and perovskite. XPS were used to measure the bottom surface of perovskite films deposited on HTLs without or with SC4A medication. The emergence of obvious S 2*p* peaks at the buried surface of SC4A modified perovskite film confirmed existence and adsorption of SC4A on the bottom surface of perovskite films (Supplementary Fig. 14), which could be due to the interaction between SC4A and perovskite. The Pb 4*f* and I 3*d* peaks shifted towards lower binding energies after SC4A modification (Fig. 3a, b), suggesting the host-guest interaction between SC4A and the Pb²⁺ and I⁻ ions in the perovskite. As exhibited in FTIR in Fig. 3c, the broadening and shift toward high wavenumber of -OH peak in SC4A+PbI₂ proved the coordination interaction between -OH in SC4A and Pb²⁺. Meanwhile, the S=O stretching vibration peak

in SC4A+PbI₂ shifted to a higher wavenumber (1163 cm⁻¹) compared with pure SC4A (1159 cm⁻¹), indicating that sulfonyl can also coordinate with Pb²⁺, thus enhancing the interaction of SC4A with perovskite. It shows that the -OH and sulfonyl can interact simultaneously with Pb²⁺, thereby passivating the undercoordinated Pb²⁺ defects at buried interface. In addition, compared with SC4A, the movement of -OH stretching vibration peak in SC4A+FAI certified the hydrogen bonding of -OH with FA⁺ and/or I⁻, and the shift of S=O stretching vibration peak affirmed the hydrogen bonding of sulfonyl with FA⁺, which is favorable for suppressing the formation of FA⁺ and I⁻ vacancies and thus inhibiting their migration (Fig. 3d). The ¹H nuclear magnetic resonance (NMR) further corroborated the above results (Fig. 3e, f). In conclusion, the -OH and sulfonyl groups in SC4A had synergistic interactions with perovskite components through multiple active sites and coordination bond and hydrogen bond, effectively passivating interface defects, restricting ions migration and improving buried interface stability.

Impact of SC4A modulation on perovskite film quality

The markedly reduced contact angle of SAM/SC4A substrate than SAM substrate resulted in improved wettability of perovskite precursor solution, which is beneficial for perovskite crystallization (Supplementary Fig. 15). Scanning electron microscopy (SEM) measurement was conducted to investigate the morphology of the perovskite films deposited on SAMs (expressed as control) and SAMs/SC4A (expressed as SC4A) substrates. Compared with control perovskite film, the SC4A perovskite film exhibited a larger grain size and smoother surface, due to ameliorated wettability and increased uniformity of SAMs (Supplementary Fig. 16). The significantly improved crystallinity was realized for the SC4A modified perovskite film as compared to the control film (Supplementary Fig. 17). Specifically, the crystallinity of the bottom surface of perovskite films was increased after incorporating SC4A (Supplementary Fig. 18). To further scrutinize the effect of SC4A on the perovskite crystallization, we used a scatheless peel-off technique to prepare samples for grazing-incidence wide-angle X-ray scattering (GIWAXS) and XRD characterizations of perovskite bottom surface (Fig. 3g and h). The scattering intensity of the (001) peak of SC4A perovskite film was higher than that of the control film, indicating that the introduction of SC4A considerably improved the crystal crystallinity of the perovskite films at the buried region. The notable improvement in the morphology and crystallinity of perovskite

bottom surface would also affect the release of residual stress. The grazing incidence X-ray Diffraction (GIXRD) technique was used to assess the residual strain at the bottom surface of perovskite films deposited on different substrates. It was obviously found that the SC4A modified perovskite film showed diminished residual strain than the control perovskite film, owing to reduced interface defects, promoted perovskite crystallization and improved buried interface morphology (Fig. 3i and Supplementary Fig. 19).

Next, the steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) were used to quantify the non-radiative recombination loss for the perovskite films deposited on HTLs substrates without and with SC4A measured from perovskite side. The carrier lifetime was prolonged from 234.45 ns of the control film to 764.25 ns of the SC4A-modulated perovskite film (Fig. 3j and Supplementary Table 1), consistent with boosted PL emission intensity (Supplementary Fig. 20), which stems from suppressed trap-assisted non-radiative recombination and improved crystallization after SC4A modification. The PL mapping test of large-area perovskite films shows that the module modified by SC4A had a more uniform PL distribution and higher PL intensity across the entire area. This means that the defects in the perovskite layer were reduced and non-radiative recombination was suppressed (Supplementary Fig. 21). The defect density of the SC4A modulated perovskite film was $0.92 \times 10^{15} \text{ cm}^{-3}$, much lower than that of the control film ($2.86 \times 10^{15} \text{ cm}^{-3}$), which is attributed to the effective passivation of the defects at the buried interface, increased grain size and improved crystallinity (Fig. 3k and Supplementary Fig. 22). Then, we measured the photoluminescence quantum yield (PLQY) and quasi-Fermi level splitting (QFLS) of the perovskite films fabricated on HTLs without or with SC4A treatment. Evidently, the SC4A-modulated perovskite film exhibited a PLQY and QFLS values of 3.82% and 1.17 eV, respectively, remarkably higher than those of the control film (2.32% and 1.14 eV), which is largely put down to substantially dropped defect density and thus inhibited trap-induced non-radiative recombination as well as reduced energy level mismatch (Fig. 3l). In brief, the quality of buried interface and perovskite films was prominently improved after SC4A manipulation, contributing to reinforced buried interface stability.

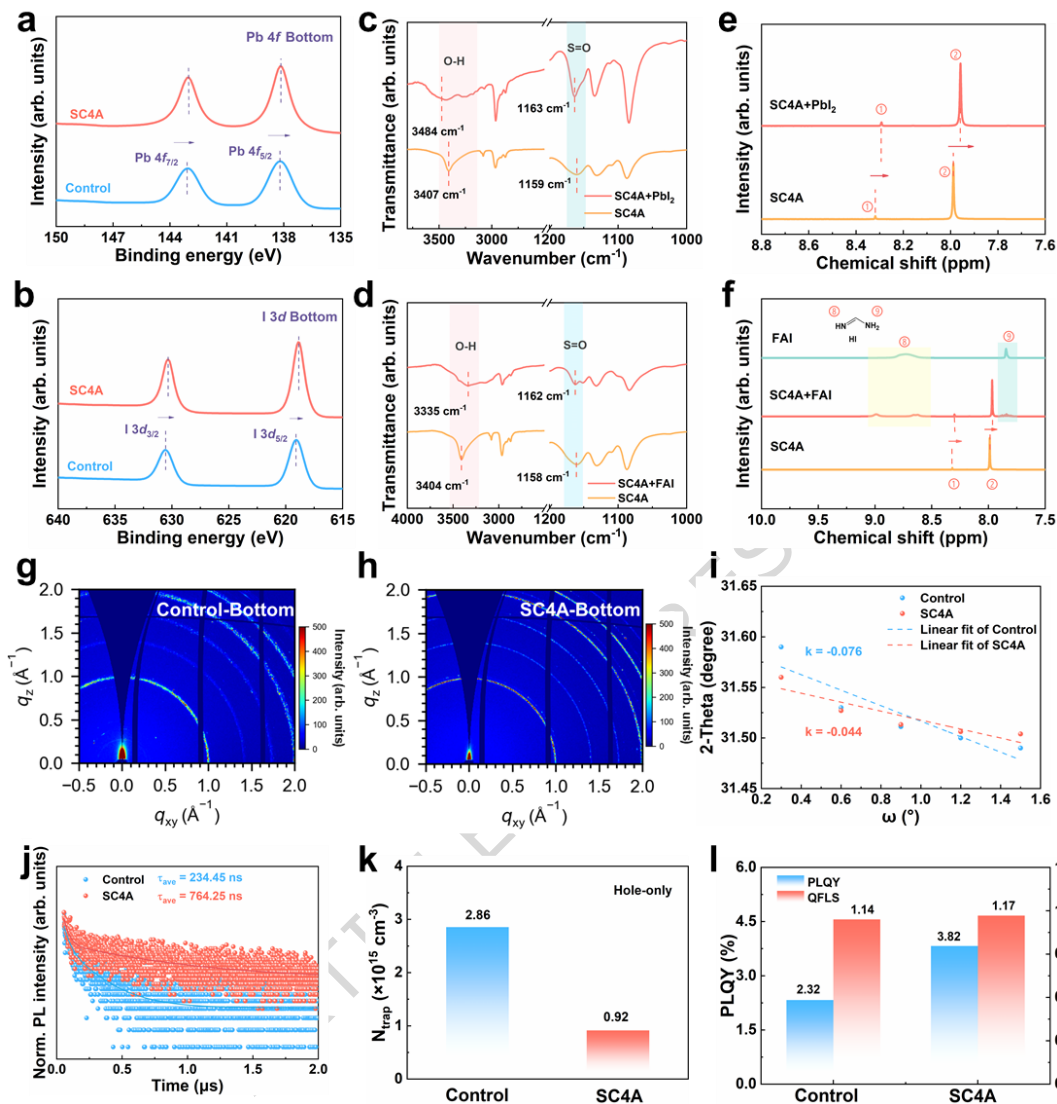


Fig. 3| Interaction mechanisms between SC4A and perovskite as well as buried interface quality. **a,b**, Pb 4f (**a**) and I 3d (**b**) XPS spectra of the bottom surfaces for the perovskite films without and with SC4A. **c**, FTIR spectra of SC4A and SC4A+PbI₂. **d**, FTIR spectra of SC4A and SC4A+FAI. **e**, ¹H NMR spectra of SC4A and SC4A+PbI₂. **f**, ¹H NMR spectra of the FAI, SC4A and SC4A+FAI. **g,h**, GIWAXS patterns of the bottom surfaces for the perovskite films without (**g**) and with (**h**). **i**, 2θ as a function of ω for the bottom surfaces for the perovskite films without and with SC4A. **j**, TRPL spectra of the perovskite films deposited on HTLs substrates without and with SC4A measured from perovskite side. **k**, Defect density for the perovskite films without and with SC4A passivation. **l**, PLQY and QFLS values of the perovskite films spin-coated on HTLs without or with SC4A.

Long-term stability of buried interface

The improved buried interface morphology, reduced interface defects and chemical bridging between SAM and perovskite via host-guest interaction should be capable of heightening the long-term stability

of buried interface. Thus, we used a scatheless peel-off technique to prepare samples for characterization of perovskite bottom surface. It needs to be stressed that the perovskite films used for characterizing the buried interface stability were aged under one sun illumination and 85 °C for 240 h. The SEM images showed that compared with the fresh control film, the fresh SC4A modified perovskite film exhibited a larger average grain size and more uniform distribution (Fig. 4a and Supplementary Fig. 23), in line with previous top surface SEM results. After 240 h of aging, we observed large cracks and voids at both surface and cross section in the control perovskite film, indicating severe surface and grain boundaries degradation (Fig. 4b). In stark contrast, the morphology of SC4A perovskite film regardless of surface and cross section was almost maintained without cracks and voids, manifesting ameliorated buried interface stability under harsh conditions. Photoluminescent properties of the bottom surface of perovskite films were examined before and after heat and light illumination stimuli (Fig. 4c). Obviously, the considerably higher and more uniform PL intensity was found in the fresh SC4A modified perovskite film compared to the fresh control film, due to minimized interfacial non-radiative recombination resulting from reduced interface defects, improved SAM homogeneity and morphology of perovskite buried surface after SC4A modulation. After 240 h of aging, the control perovskite film underwent faster PL intensity decay than the SC4A-modulated film.

KPFM measurements were subsequently conducted to investigate the potential variations of the bottom surface of perovskite films without and with SC4A (Supplementary Fig. 24). Notably, the CPD of control sample was decreased from 432 mV to 314 mV after 240 h aging, indicating substantial perovskite decomposition (Fig. 4d). In comparison, the SC4A sample exhibited a considerably smaller change, benefiting from suppressed perovskite decomposition after introducing SC4A. Furthermore, the FWHM of the CPD values on the SC4A modified film surface was narrowed regardless of fresh and aged samples, indicating the enhanced potential uniformity. The XRD results showed that the characteristic peaks of PbI_2 and $\delta\text{-FAPbI}_3$ emerged in the control perovskite film after 120 h of aging, whereas no characteristic peaks of PbI_2 and $\delta\text{-FAPbI}_3$ was observed in the SC4A modified film even if after 240 h, once again corroborating saliently inhibited perovskite decomposition following SC4A manipulation (Fig. 4e, f). In a word, benefiting from supramolecular host-guest interaction of SC4A, the suppressed SAM aggregation, enhanced SAM uniformity, reduced nanovoids and defects of perovskite bottom surface, as well as

chemical bridging between SAM and perovskite should be primarily responsible for reinforced buried interface stability, ultimately enhancing the operational stability of inverted PSCs.

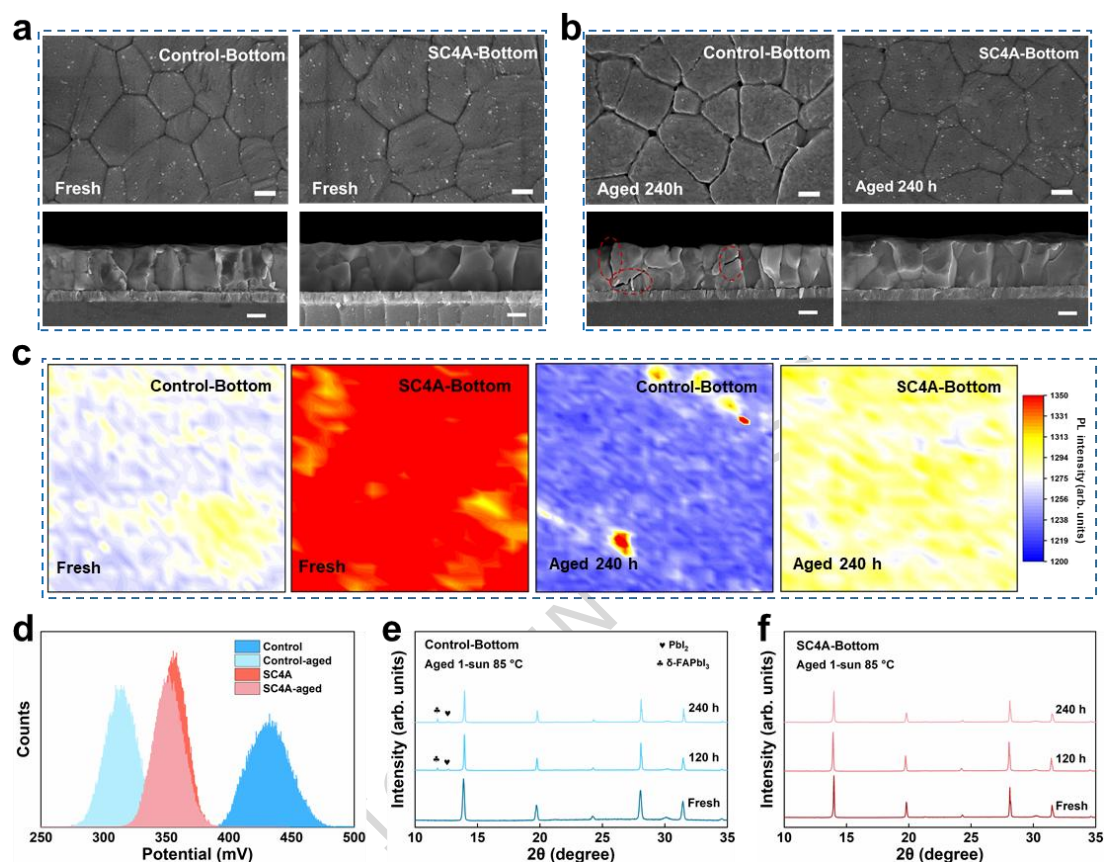


Fig. 4| Effect of SC4A on buried interface stability and quality. **a**, SEM images of the fresh perovskite bottom surface without and with SC4A modification (Surface scale bars are 200 nm, and cross-section scale bars are 300 nm). **b**, SEM images of the bottom surfaces for the perovskite films without or with SC4A treatment after aging under 85 °C and 1-sun equivalent illumination for 240 h (Surface scale bars are 200 nm, and cross-section scale bars are 300 nm). **c**, PL mapping of the bottom surfaces for the perovskite films without or with SC4A before and after aging under 85 °C and 1-sun equivalent illumination for 240 h. **d**, CPD distribution histogram derived from the KPFM of the bottom surfaces for the perovskite films without or with SC4A before and after aging under 85 °C and 1-sun equivalent illumination for 240 h. **e**, **f**, XRD patterns of the bottom surfaces for the perovskite films without (**e**) or with (**f**) SC4A after aging under 85 °C and 1-sun equivalent illumination for different times.

Photovoltaic performance

The inverted PSCs based on C4A, TC4A and SC4A modification were fabricated using antisolvent method. As expected, the photovoltaic performance was enhanced for all supramolecules but SC4A-based devices exhibited the best photovoltaic performance (Supplementary Fig. 25). After SC4A regulation, the PCE was significantly increased from 23.99% to 26.85%, due to improvements in all photovoltaic

parameters (Supplementary Fig. 26). The external quantum efficiency (EQE) results in Supplementary Fig. 27 show that the SC4A-based device had the highest EQE in the visible light range, confirming the reliability of the J - V measurement. To further enhance PCE and demonstrate universality of our SC4A modification method for perovskite film preparation approaches, vacuum flash evaporation technique was used to fabricate small- and large-area PSCs. In view of this, we further evaluated the effect of SC4A modulation on photovoltaic performance of 1.53 eV, 1.68 eV and 1.78 eV inverted PSCs. Based on vacuum flash evaporation method, the champion control device delivered a PCE of 25.16%, with a short-circuit current density (J_{sc}) of 25.90 mA/cm², an open-circuit voltage (V_{oc}) of 1.164 V, and a fill factor (FF) of 83.44% (Fig. 5a, Supplementary Fig. 28 and Supplementary Table 2). By comparison, the SC4A modulated device yielded an optimal efficiency of 27.36%, corresponding to a J_{sc} of 26.17 mA/cm², a V_{oc} of 1.198 V, and an FF of 87.26%. Moreover, the SC4A modulated device accomplished a certified stabilized PCE of 27.12% (Supplementary Fig. 29), which is one of the highest certified PCEs ever reported for single-junction PSCs (Fig. 5b and Supplementary Table 3). The histogram of PCEs shown in Supplementary Fig. 30 confirmed good reproducibility of the PCE improvement with SC4A modification. The SC4A buried interface modulation method was also extended to 1.68 eV and 1.78 eV wide-bandgap PSCs, which is a critical sub-cell for tandem devices, proving universality of our interface modification method. For 1.68 eV PSCs, the PCE was improved from 22.38% to 24.39% after SC4A modification (Supplementary Fig. 31 and Supplementary Table 4), which is among the highest PCEs reported for 1.66-1.68 eV PSCs so far (Supplementary Fig. 32 and Supplementary Table 5). In addition, the SC4A-modified 1.78 eV PSCs fulfilled a maximum PCE of 20.30%, much higher than 18.69% of the control device (Supplementary Fig. 33 and Supplementary Table 6), which represents one of the highest PCEs reported for 1.78-1.80 eV PSCs up to now (Supplementary Fig. 34 and Supplementary Table 7). Subsequently, we explored the application of SC4A in all-perovskite tandem solar cells. The optimized tandem device achieved a PCE of 28.17% (Supplementary Fig. 35 and Supplementary Table 8). The results further demonstrate its broad applicability in tandem solar cells. We also considered the upscaling capability and successfully fabricated high-performance modules (30 cm×30 cm; Supplementary Fig. 36). After SC4A manipulation, the PCE of perovskite solar modules (PSMs) was enhanced from 20.11% to 22.39% (Supplementary Fig. 37), corroborating upscaling ability of our soft supramolecule interface modulation

method. The SC4A modified PSM with an aperture area of 655.2 cm² achieved a certified PCE of 22.25% (Fig. 5c and Supplementary Fig. 38), which is among the highest PCEs ever reported for 500~1000 cm² PSMs (Fig. 5d and Supplementary Table 9).

Interfacial carrier dynamics

Interfacial carrier dynamics was investigated to uncover the origins of improved photovoltaic performance following SC4A modulation. As presented in Supplementary Fig. 39, the Mott-Schottky analysis showed that the built-in potential (V_{bi}) was considerably increased from 0.87 V of the control device to 1.12 V of SC4A-treated device, which is principally attributed to the optimized energy level alignment and reduced interfacial non-radiative recombination due to SC4A incorporation. Then, the hole extraction efficiency at buried interface was assessed by performing transient reflection spectroscopy (TRS) measurements. The surface hole extraction velocity (S) value of HTL/SC4A/perovskite sample was 2850±80 cm/s, higher than that of the HTL/perovskite sample (2340±150 cm/s), demonstrating the improved hole extraction capability after SC4A modulation (Fig. 5e, f). Based on the TRPL results from the glass/substrate side (Supplementary Figs. 40 and 41 and Supplementary Table 11), the average carrier lifetime (τ_{ave}) was decreased from 119.63 ns to 94.87 ns after SC4A modification, indicating more efficient extraction of photo-generated holes into hole transport layer. From transient photocurrent (TPC) in Supplementary Fig. 42, the photocurrent decay lifetime was shortened for SC4A-modified device compared with the control device, further confirming promoted hole extraction and transfer at buried interface (Supplementary Table 12). The SC4A-modified device exhibited much longer photovoltage decay lifetime than the control device (301.84 μ s VS 123.14 μ s), which is suggestive of markedly suppressed interfacial non-radiative recombination after incorporating SC4A (Supplementary Fig. 43 and Supplementary Table 13). The carrier recombination process under light illumination was further evaluated by light intensity dependent V_{oc} (Supplementary Fig. 44). The ideality factor of the control device was 1.90, which is much larger than 1.53 of the SC4A-modified devices, indicative of substantially suppressed trap-assisted non-radiative recombination upon introducing SC4A. Besides, compared with the control device, the lower dark-state saturation current density of the SC4A-modified device further verified inhibited non-radiative recombination (Supplementary Fig. 45). As revealed by electrochemical

impedance spectroscopy (EIS), the SC4A-modified device had a much larger recombination resistance than the control device, indicating prominently restrained non-radiative recombination (Supplementary Fig. 46). It can be concluded that the mitigated interface defects, improved energy level alignment, and ameliorated buried interface contact should jointly account for significantly reduced non-radiative recombination loss, thereby contributing to enhanced photovoltaic performance.

Device stability

Stability remains a long-standing challenge in perovskite photovoltaic technology. Thus, effect of the SC4A buried interface modification on device stability was estimated. We performed the MPP tracking for the control and SC4A-modulated inverted PSCs under continuous white LED illumination at 100 mW/cm². After 2125 h of aging, the SC4A-modulated device maintained 97.4% of its initial PCE, while the control device degraded to 88.8% of its original PCE, suggesting improved operational stability upon SC4A modification (Fig. 5g). In addition, according to the IEC 61215 standard, the encapsulated control and SC4A-modulated devices were subjected to damp heat test under 85 °C and 85% RH. As shown in Fig. 5h, the SC4A modified device retained 90.7% of its initial efficiency after 2000 h of aging, whereas the control device dropped to 71.9% of its initial PCE, indicating enhanced damp heat longevity after the introduction of SC4A. Finally, the improved long-term ambient stability was realized for the SC4A modified device compared with the control device (Supplementary Fig. 47). The reduced interface defects, chemical bridging and improved buried interface stability should be main reasons for increased device durability. Therefore, our supramolecular host-guest interaction strategy based on soft SC4A supermolecule conferred inverted PSCs both high efficiency and exceptional operational stability, which propelled the development of inverted PSCs towards practical deployment.

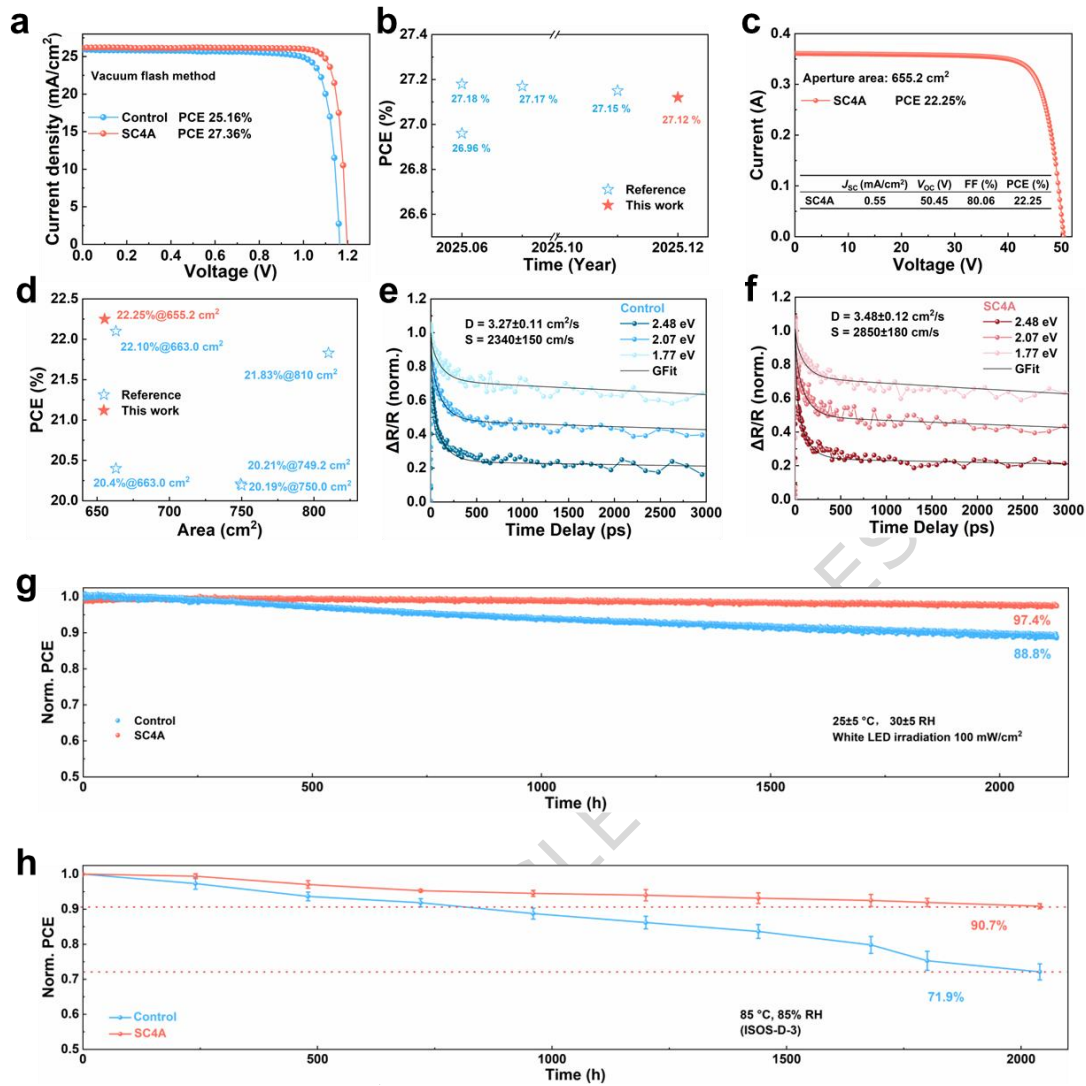


Fig. 5 | Photovoltaic performance of PSCs. **a**, J - V curves of the champion 1.53 eV control and SC4A PSCs based on vacuum flash evaporation. **b**, Comparison of the certification PCE of our PSCs with those of other reported PSCs with PCEs $\geq 27\%$. **c**, J - V curves of the certified champion modules modulated by SC4A. **d**, Summary of reported efficient 500-1000 cm² PSC modules with PCEs surpassing 20%. **e, f**, TRS for the perovskite films deposited on HTLs without or with SC4A probed at 2.48, 2.07, and 1.77 eV. **g**, Operational stability of the encapsulated devices without or with SC4A under continuous 1-sun illumination at MPP (The initial PCE of the control and SC4A-modified devices is 25.88% and 27.14%, respectively). **h**, Damp heat stability of the encapsulated devices without or with SC4A aged under 85 °C and 85% relative humidity following the IEC61215:2016 standard (The initial highest PCE of the control device is 24.89% and the average PCE is 24.49%, while the initial highest PCE the SC4A-modified device is 25.96% and the average PCE is 25.74%). Error bars represent the standard deviation of multiple independent measurements.

Discussion

In summary, we stabilized buried interface of inverted PSCs by a supramolecular host-guest interaction strategy based on soft SC4A. SC4A modification can suppress SAMs aggregation, promote homogeneous distribution of SAMs, passivate interface defects, release interface residual stress, reduce interface nanovoids, as well as improve stability of buried interface, by simultaneous chemical interaction with perovskites and SAMs via hydrogen bonding, coordination bonding and van der Waals forces. Ultimately, the SC4A-modulated PSCs and large-area modules fulfilled a certified PCE of 27.12% and 22.25% (655.2 cm^2), respectively, which both are among the most sophisticated inverted PSCs and modules. Moreover, SC4A modification resulted in reinforced buried interface stability, which was translated into prominently enhanced device stability, maintaining 97.4% of its initial PCE after continuous MPPT under one sun illumination for 2125 h, and 90.7% of its original PCE after damp heat aging (85 °C and 85% relative humidity) for 2000 h. This work provides insights into addressing the instability of buried interface through supramolecule engineering from the perspective of function-driven molecular design, which paves the way for the commercial application of perovskite photovoltaics.

Methods

Materials

All reagents were used directly without further purification. 4-tert-Butylcalix[4]arene (C4A, 98%), 4-tert-Butylthiacalix[4]arene (TC4A, 98%) and 4-tert-Butylsulfonylcalix[4]arene (SC4A, 98%) were purchased from Macklin Biochemical Co., Ltd. Caesium iodide (CsI, 99.99%), methylazanium, bromide (MABr, 99.99%), lead (II) iodide (PbI_2 , 99.99%) and lead(II) thiocyanate ($\text{Pb}(\text{SCN})_2$, 99.99%) were obtained from Xi'an Yuri Solar Co., Ltd. Bathocuproine (BCP, 99.9%), formamidinium iodide (FAI, 99.9%), lead (II) bromide (PbBr_2 , 99.99%), methylammonium iodide (MAI, 99.99%), [6,6]-phenyl-C₆₁-butyric acid methyl ester (PC₆₁BM, 99.99%) were bought from Advanced Election Technology Co., Ltd. [4-(3, 6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz, 99%) and [2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz, 98%) were purchased from TCI America. N,N-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), chlorobenzene (CB, 99.8%), isopropanol (IPA, 99.7%) and ethanol were acquired purchased from Sigma Aldrich Co., Ltd. NiO_x nanoparticles were synthesized by the chemical precipitation method. ITO substrates were purchased from

Beijing Huamin New Materials Technology Co., Ltd.

Preparation of 1.53 eV bandgap inverted PSCs based on antisolvent method

The ITO (1.49 cm×1.49 cm) was successively washed with detergent solution, deionized water and absolute ethanol under sonication for 15 min, followed by drying with a pure nitrogen flow. The cleaned ITO substrates were subjected to UV-ozone treatment for 15 min. Subsequently, a NiO_x NPs dispersion (25 mg/mL in deionized water) was spin-coated on the ITO substrates at 5000 rpm for 30 s. After that, the NiO_x films were annealed at 150 °C for 10 min in air, and then transferred into a glovebox filled with nitrogen. A mixed solution of Me-4PACz and MeO-2PACz (0.3 mg/mL in ethanol, with a mass ratio of 1:1) was spin-coated on the NiO_x-covered substrate at 3000 rpm for 30 s and then annealed at 100 °C for 10 min. For the devices with interface modification, different concentrations of C4A, TC4A or SC4A solutions (0.5, 0.8, 1.0, and 1.2 mg/mL, dissolved in CB:DMF = 1:1) were spin-coated on the SAMs layer at a speed of 5,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. For the preparation of perovskite film with a composition of FA_{0.95}CS_{0.05}PbI₃, 1.5 M perovskite precursor solution was prepared by dissolving 245.06 mg FAI, 19.49 mg CsI, and 691.5 mg PbI₂ in a mixed solvent of DMF and DMSO with a volume ratio of 4:1. The perovskite precursor solution was spin-coated on the HTL substrates without or with SC4A treatment at 2,000 rpm for 10 s and 4,000 rpm for 40 s in a nitrogen glovebox. Then, 150 µL of CB was dropped quickly at 7 s before the end of the spin-coating process. Finally, the perovskite film was annealed at 100 °C for 30 min. After cooling to room temperature, PEAi (1.8 mg/mL in IPA) solution was spin-coated on top of the as-prepared perovskite films at 5,000 rpm for 30 s, and annealed at 100 °C for 5 min. Subsequently, PC₆₁BM (23 mg/mL in CB) solution was spin-coated onto the PEAi-treated perovskite film at 2,500 rpm for 30 s. Next, the BCP solution (0.5 mg/mL in IPA) was spin-coated on the PC₆₁BM layer at 5,000 rpm for 30 s. Finally, the device was completed by thermally evaporating the electrode of 100 nm Ag.

Fabrication of 1.53 eV bandgap inverted PSCs based on vacuum flash evaporation method

The FTO substates were successively washed by detergent, deionized water, acetone and ethyl alcohol for 20 min. Then, the FTO substates were dried by nitrogen, followed by ultraviolet ozone treatment for 30 min. Then, the clean FTO was transferred into a glovebox filled with nitrogen. The 4PADCB solution (0.5 mg/mL in ethanol) was spin-coated onto the FTO substrate at 3000 rpm for 30 s and annealed at 100 °C

for 10 min. For SC4A-modulated device, the SC4A solution (0.8 mg/mL, dissolved in CB:DMF = 1:1) was spin-coated on the SAMs layer at a speed of 5000 rpm for 30 s, followed by annealing at 100 °C for 5 min. For the preparation of perovskite film with a composition of $\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.9}\text{PbI}_3$, 1.82 M perovskite precursor solution was prepared by dissolving 23.64 mg CsI, 297.22 mg FAI, 12.29 mg MAI, and 880.97 mg of PbI_2 in a mixed solvent of DMF and DMSO with a volume ratio of 4:1. The perovskite precursor solution was spin-coated on the substrate at 1,000 rpm for 10 s and 3,000 rpm for 30 s. Then, the wet perovskite film was promptly placed into a sample chamber that was connected to vacuum-pumping equipment. By opening the valve that linked the specimen chamber to the pump system, the perovskite film was instantly exposed to a low pressure of 10 Pa and maintained for 30 s. After that, the specimen chamber was fully pressurized by allowing ambient air to enter. Subsequently, the film was annealed at 120 °C for 15 min in an environment with a relative humidity (RH) of $30 \pm 10\%$. After cooling to room temperature, PDI_2 solution (0.4 mg/mL in IPA) was spin-coated onto the perovskite film at 5,000 rpm for 30 s and annealed at 100 °C for 5 min. Then, 30 nm C_{60} and 7 nm BCP were thermally evaporated in sequence. Finally, a 100 nm silver electrode was thermally evaporated on the top of BCP film.

Fabrication of 1.68 eV bandgap inverted PSCs using antisolvent method

The ITO substrates were successively washed by detergent, deionized water, acetone and ethyl alcohol for 20 min. Then, the ITO substrates were dried by nitrogen, followed by ultraviolet ozone treatment for 30 min. NiO_x nanoparticles were dispersed in a mixed solvent of deionized water and isopropanol (IPA) with a volume ratio of 3:1 to obtain a 10 mg/mL suspension. The NiO_x layer was deposited on pre-cleaned ITO substrates by spin-coating at 2000 rpm for 30 s and subsequently annealed at 150 °C for 10 min. Then, the substrates were immediately transferred into a nitrogen-filled glovebox. Next, 0.3 mg/mL Me-4PACz ethanol solution was spin-coated onto the NiO_x layer at 3000 rpm for 30 s to form SAM layer, followed by annealing at 100 °C for 10 min, completing the HTL fabrication. 80 μL of perovskite precursor solution was dispensed onto the substrate and spin-coated using a two-step program: 1,000 rpm for 5 s (500 rpm s^{-1} acceleration) followed by 5,000 rpm for 30 s (2500 rpm s^{-1} acceleration). During the second stage, 200 μL of CB was dripped on the perovskite film at the final 20 s. The wet perovskite film was immediately annealed at 100 °C for 20 min. After cooling to room temperature, the interface modification layer was deposited by spin-coating a solution of EDAI_2 (1 mg/mL in IPA, pre-stirred at 70 °C for >3 h), followed

by annealing at 100 °C for 5 min. Subsequently, 23 mg/mL PC₆₁BM solution (in CB) was spin-coated at 2,500 rpm for 40 s and annealed at 70 °C for 10 min. Then, 0.5 mg/mL BCP solution was deposited on PC₆₁BM film at 5000 rpm for 30 s. Finally, 100 nm Ag electrode was thermally evaporated on BCP film to complete the device fabrication.

Preparation of 1.78 eV bandgap inverted PSCs employing antisolvent method

The cleaned ITO substrates were subjected to UV-ozone treatment for 15 min. Subsequently, a NiO_x NPs dispersion (20 mg/mL in water) was spin-coated on the ITO substrates at 2000 rpm for 60 s, annealed at 150 °C for 20 min in air, and then transferred into a nitrogen glovebox. The Me-4PACz solution (0.33 mg/mL in ethanol) was spin-coated onto the NiO_x substrate at 4,000 rpm for 40 s and annealed at 100 °C for 10 min. For SC4A-modulated device, the SC4A solution (0.8 mg/mL, dissolved in CB:DMF = 1:1) was spin-coated on the SAMs layer at a speed of 5,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. For the preparation of wide-bandgap perovskite films with a composition of Cs_{0.2}FA_{0.8}Pb(I_{0.6}Br_{0.4}), 1.2 M perovskite precursor solution was prepared by dissolving 165.09 mg FAI, 62.35 mg CsI, 264.25 mg PbBr₂, 7.76 mg Pb(SCN)₂ and 221.28 mg PbI₂ in a mixed solvent of DMF and DMSO with a volume ratio of 3:1. The perovskite precursor solution was spin-coated on the substrate at 500 rpm for 2 s and 4,000 rpm for 60 s in a nitrogen glovebox. Then, 700 µL of diethyl ether was rapidly dripped at 35 s before the end of the spin-coating process. Finally, the perovskite film was annealed at 100 °C for 15 min. Then, the PEAi solution (1.8 mg/mL in IPA) was spin-coated on top of the as-prepared perovskite films at 3,000 rpm for 30 s, and annealed at 100 °C for 2 min. Following cooling, the PC₆₁BM solution (23 mg/mL in CB) was spin-coated onto the perovskite film at 2,500 rpm for 30 s. After that, BCP solution (0.5 mg/mL in IPA) was spin-coated on the PC₆₁BM layer at 5,000 rpm for 30 s. Ultimately, 100 nm Ag was thermally evaporated on the surface of BCP films.

Preparation of all-perovskite tandem solar cells

For the WBG sub-cell, a 1.78 eV bandgap inverted PSCs preparation method was adopted. After ALD-SnO₂ deposition, the substrates were transferred back to the thermal evaporation system to deposit an ultra-thin layer of Au clusters (about 1 nm) on ALD-SnO₂. PEDOT:PSS solution was spin-coated on top of front cells at 4,000 rpm for 30 s and then annealed at 120 °C for 20 min. Then, the substrates were immediately transferred into a nitrogen-filled glovebox for the deposition of NBG perovskite films. For the preparation of perovskite film with a composition of Cs_{0.1}FA_{0.6}MA_{0.3}Pb_{0.5}Sn_{0.5}I₃, 2.0 M perovskite

precursor solution was prepared by dissolving 51.96 mg CsI, 95.38 mg MAI, FAI 206.36 mg, SnI₂ 372.51 mg, PbI₂ 461.01 mg, SnF₂ 11.75 mg in a mixed solvent of DMF and DMSO with a volume ratio of 3:1 and shook overnight. 50 μ L of perovskite precursor solution was dispensed onto the substrate and spin-coated using a two-step program: 1000 rpm for 5 s followed by 4000 rpm for 50 s. During the second stage, 300 μ L of CB was dripped on the perovskite film at the final 15 s. The wet perovskite film was immediately annealed at 100 °C for 5 min. Then, the EDAl₂ solution (0.5 mg/mL in IPA) was spin-coated on top of the as-prepared perovskite films at 4,000 rpm for 25 s, and annealed at 100 °C for 5 min. Following cooling, 15 nm C₆₀ and 5 nm BCP were thermally evaporated in sequence. Finally, a 100 nm silver electrode was thermally evaporated on the top of BCP film.

Module fabrication

For the 300 $\times$ 300 mm² perovskite submodule, series-connected 42 sub-cells were fabricated by sequential laser scribing of P1, P2, and P3 lines. The P1 lines were patterned using a 1064 nm laser, while the P2 and P3 lines were formed using a 532 nm laser, resulting in a geometric fill factor of 98%. It needs to be stressed that SAM, SC4A and perovskite films were fabricated in the air (RH 20~25%, 22 °C). After the P1 scribing process, the ITO-coated glass substrates were thoroughly cleaned, followed by the deposition of a NiO_x layer via magnetron sputtering. Subsequently, a mixed solution of Me-4PACz and MeO-2PACz (0.3 mg/mL in ethanol, with a mass ratio of 1:1) was slot-die coated and annealed at 100 °C for 10 min. The interface modification layer was deposited on top of the SAMs layer by slot-die coating the SC4A solution (0.8 mg/mL, dissolved in CB:DMF = 1:1), followed by annealing at 100 °C for 5 min. The coating speed was 10 mm/s, and the gap between the substrate and the slot-die head was maintained at approximately 60 μ m. The perovskite precursor solution was deposited on the SC4A-treated HTL substrates by slot-die coating at room temperature. The coating speed was 8 mm/s, and the gap between the substrate and the slot-die head was maintained at approximately 50 μ m. Immediately after coating, the perovskite wet films were rapidly transferred into a vacuum drying chamber, where the pressure was reduced to below 10 Pa within 10 s and maintained under reduced pressure for 10-110 s. The resulting perovskite films were then annealed at 100 °C for 30 min in ambient air. Subsequently, LiF and C₆₀ were sequentially deposited onto the perovskite films by thermal evaporation, followed by the deposition of SnO_x via atomic layer deposition. After these functional layers were deposited, P2 lines were patterned

by laser scribing. Then, 20 nm ITO and 80 nm Cu layers were sequentially deposited by magnetron sputtering. Finally, P3 lines were fabricated to complete the series interconnection of the sub-cells.

Characterizations of PSCs

The J - V curves of the devices were evaluated at room temperature under standard illumination conditions using a Keysight B2901BL source meter coupled with a solar simulator featuring a 450 W xenon lamp, providing irradiance equivalent to 1 sun. Light intensity was precisely adjusted to match the AM 1.5G spectrum (100 mW/cm^2) by calibrating with a reference silicon photovoltaic cell. A black metal aperture mask was used to define device active area as 0.053 cm^2 for devices using vacuum flash evaporation method and 0.1 cm^2 for the devices using antisolvents. EQE spectra were acquired across the 300-900 nm wavelength range using a Newport-74125 instrument system, employing a 300 W xenon lamp as the excitation source.

Stability tests of solar cells

For MPPT stability, the YH-VMPP-16 MPP tracking system was used (Suzhou Yanghua) with the white LED light source (approximately 100 mW/cm^2) to evaluate the operational stability of the sealed devices at MPP under continuous one-sun illumination conditions.

For damp heat stability, the encapsulated devices were subjected to a damp heat (DH) test in a programmable constant temperature and humidity test chamber (HS-408L) under the conditions of $85 \pm 0.5 \text{ }^\circ\text{C}$ and $85 \pm 0.5\%$ relative humidity (RH). After measuring the initial efficiency of the devices, they were placed in the test chamber. After the equipment reached 85% RH and $85 \text{ }^\circ\text{C}$ again, the measurement started.

Certification of PSCs

For the certification of small-area PSCs, the PSC devices were sent to the Tianjin Institute of Metrological Supervision and Testing, China, for official certification. Specifically, we filled the prepared solar cell samples for certification testing with nitrogen, then stored them with a vacuum sealer and always placed them in a dark sealed box. During the certification test, the certification agency recalibrated the effective active area of our solar cells to ensure accurate evaluation of data such as current density.

For the certification of the perovskite solar module, one of high-efficiency SC4A-modulated perovskite

solar modules was sent to the TUV SUD, Jiangsu Province, China, for official certification. The encapsulated modules were subjected to light-soaking and then tested continuously. The measured aperture area of module was 655.2 cm². All tests were performed at about 25 °C and a relative humidity of approximately 40%.

Fourier transform infrared spectroscopy measurements

FTIR was recorded on a Thermo-Fisher Nicolet iS50 Infrared Fourier transform microscope. Raman spectroscopy was performed using the Renishaw in Via system (United Kingdom).

X-ray photoelectron spectra and ultraviolet photoelectron spectra measurements

The XPS and UPS measurements were carried out on a Thermo-Fisher ESCALab 250Xi system with a monochromatized Al K α (for XPS mode) under the pressure of 5.0×10^{-7} Pa.

UV-vis absorption spectra measurements

UV-vis absorption spectra were collected using a UV-vis spectrometer (Shimadzu UV-3600, Japan).

X-ray diffraction and grazing incidence X-ray diffraction measurements

The XRD patterns were measured by Rigaku Ultima IV (Japan). The GIXRD patterns were collected by PANalytical Empyrean (Netherlands). The GIXRD technique utilized the 2θ - $\sin^2\psi$ method for linear regression analysis to determine residual strain, and the corresponding calculation formula is as follows:

$$\varepsilon_{\psi} = \frac{d_{\psi} - d_0}{d_0} = \frac{1}{2} \left(\cot\theta_0 \cdot \frac{\Delta 2\theta_{\psi}}{\sin^2\psi} \right) \quad (1)$$

where ψ signifies sample tilt angle, ε_{ψ} represents strain, d_{ψ} signifies the interplanar spacing under strained conditions, d_0 represents the interplanar spacing in the unstrained state, and θ denotes the Bragg angle.

Scanning electron microscope measurements

SEM measurements were conducted by employing a secondary electron scanning electron microscopy (ZEISS Sigma 300, Germany).

Atomic force microscope and Kelvin probe force microscope measurements

The AFM and KPFM measurements were carried out by a Bruker Dimension Icon system (Germany).

Electrochemical impedance spectroscopy and Mott-Schottky plot measurements

The EIS and Mott-Schottky plot measurements were conducted through a CHI660F electrochemical workstation under ambient conditions.

Steady-state photoluminescence and time-resolved photoluminescence measurements

Steady-state and time-resolved PL spectra were recorded by spectrofluorometer (Edinburgh FLS980, UK) using a 460 nm pulsed laser. PL mapping was collected via Laser Ramon Instrument. The dual-exponential model was applied to fit TRPL data, allowing for the determination of the average carrier lifetime (τ_{ave}). The τ_{ave} was calculated based on the following formula:

$$\tau_{ave} = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2} \quad (2)$$

where A_1 and A_2 represent the amplitudes of the fast and slow decay components, respectively, reflecting the relative contributions of distinct decay pathways. τ_1 and τ_2 denote the lifetimes associated with fast and slow decay processes, respectively, corresponding to non-radiative recombination at the interface and radiative recombination in the bulk material.

Space charge limited current measurements

The hole-only device with a structure of ITO/NiO_x/SAM/perovskite/PTAA/Ag was subjected to SCLC characterization. The calculation formula for the defect density (N_t) in perovskite films is described as follows:

$$N_t = \frac{2\varepsilon_0\varepsilon V_{TFL}}{eL^2} \quad (3)$$

Where ε_0 is the vacuum permittivity (8.8542×10^{-14} F/cm), ε is the relative permittivity of the perovskite, V_{TFL} is the starting voltage of trap filled limit voltage, e is the elementary charge, and L is the thickness of the active layer.

Photoluminescence quantum efficiency and Quasi-Fermi level splitting measurements

The measurements of PLQY and QFLS under varying light intensities were conducted using the Lumin QY quasi-fermi level splitting detection system developed by ideaoptics Inc. The absolute fluorescence emission spectra were acquired through a fiber optic spectrometer integrated with a 100 mm diameter integrating sphere. The excitation source employed a 785 nm laser with a 5 mm diameter excitation spot. The spectral reception system was calibrated using a broadband halogen light source certified by the National Institute of Metrology, China. The computational formula for QFLS is expressed as follows:

$$QFLS = k_B T \cdot \ln \left(PLQY \frac{J_G}{J_{0,rad}} \right) \quad (4)$$

where k_B is the Boltzmann constant. T is the temperature (300 K). J_G is the generation current density under illumination, in this case, approximated to the J_{SC} of devices. $J_{0,rad}$ is the dark radiative recombination saturation current density.

Nuclear magnetic resonance spectroscopy measurements

The liquid-state nuclear magnetic resonance (NMR) measurements were performed in d_6 -DMSO solvent using a Bruker 400 MHz spectrometer.

Grazing-incidence wide-angle X-ray scattering measurements

The GIWAXS measurements were performed at the BL03HB beamline of the Shanghai Synchrotron Radiation Facility (SSRF), China (data processed by GIWAXS-Tools).

The Transient reflection spectroscopy measurements

Transient reflection (TR) measurements were performed by a pump-probe spectrometer (TA-100, Time-tech spectra). A Ti:Sapphire amplifier (Astrella, Coherent) was used to generate 800 nm light at 1 kHz repetition rate. The fundamental pulse was split into two parts. One part was sent to an optical parametric amplifier (TOPAS, Light conversion) for the various pump wavelength generation at 1.65 eV, 2.1 eV and 2.6 eV. The pump was chopped at a frequency of 500 Hz and attenuated by neutral density filter wheels. The other part of the fundamental pulse was focused into a sapphire crystal to generate a visible continuum (450-810 nm) that is used as the probe. The probe pulses were delayed in time with respect to the pump pulses using a motorized translation stage mounted with a retroreflecting mirror. The pump and probe were spatially overlapped on the surface of the sample. For TR measurement, the incident angle for pump was around 0° and probe was around 45° . The initial carrier density is on the order of 10^{16} cm^{-3} .

Transient reflection kinetics modelling: The measured surface carrier kinetics can be modelled with diffusion and surface recombination/transfer model. The carriers are generated with profile

$$N(x,0)=N_0e^{-\alpha x} \quad (5)$$

where N_0 is the initial surface carrier density. If the traces are normalized, then N_0 is equal to 1. The values for the absorption coefficient at different pump-photon energies for samples are known. The carrier movement obeys the diffusion and surface recombination/transfer equation:

$$\frac{\partial N(x,t)}{\partial t} = D \frac{\partial^2 N(x,t)}{\partial x^2} - \frac{N(x,t)}{\tau_B} \quad (6)$$

$$\begin{aligned}\frac{\partial N(x,t)}{\partial t}\bigg|_{x=0} &= \frac{S_F}{D} N(0,t) \\ \frac{\partial N(x,t)}{\partial t}\bigg|_{x=L} &= -\frac{S_L}{D} N(L,t)\end{aligned}$$

where $N(x, t)$ is the carrier density as a function of depth (x) and time (t), D is the ambipolar diffusion coefficient and τ_B is the bulk carrier lifetime. Generally, in perovskite material, τ_B is on the order of hundreds of ns, in the 5000 ps probe window, it can be neglected. The boundary condition can be described by the followings, for the front surface, surface recombination/transfer will dominate (SF). For the back surface, surface recombination/transfer happens with SL. For pure perovskites, the back surface and front surface can be assumed to have the same surface recombination velocity. For samples with HTL, the front surface transfer velocity is different from the back where the back (SL) takes the value of pure perovskites. L is the thickness of the film.

Density functional theory calculations

First-principles calculations were performed using density functional theory (DFT). The electron-ion interactions were described by the projector augmented-wave (PAW) method, and the exchange-correlation functional was treated within the generalized gradient approximation (GGA) parameterized by Perdew, Burke, and Ernzerhof (PBE). A plane-wave basis set with a kinetic energy cutoff of 400 eV was adopted for all calculations. The FAPbI₃ surface model was constructed as a periodic slab with five atomic layers, separated by a vacuum layer of ~ 30 Å perpendicular to the surface to prevent artificial interactions between periodic images. During structural optimization, the bottom two layers were fixed at their bulk-truncated positions, while the upper three layers (and adsorbed molecules) were fully relaxed. The NiO surface was modeled as a four-atomic-layer slab with a 30 Å vacuum layer. Spin-polarization was explicitly included for NiO calculations. To account for strongly correlated Ni 3d electrons, the GGA+ U method (Dudarev formulation) was used with an effective Hubbard U ($U_{\text{eff}} = 8.0$ eV) on Ni atoms. During geometry optimization, the bottom layer was fixed, while the remaining layers and adsorbates relaxed. Van der Waals (vdW) interactions, critical for molecular adsorption, were included using Grimme's DFT-D3 method with Becke-Johnson damping. A dipole correction was applied normal to the surface to eliminate spurious electrostatic interactions from periodic boundary conditions. Convergence criteria were set to 1×10^{-4} eV (electronic SCF) and 0.03 eV/Å (ionic relaxation). The binding energy (E_b) was calculated as $E_b = E(\text{surface@molecule}) - E(\text{surface}) - E(\text{molecule})$, where

$E(\text{surface@molecule})$ is the energy of the optimized adsorption system, $E(\text{surface})$ is the energy of the relaxed clean surface slab, and $E(\text{molecule})$ is the energy of an isolated molecule calculated in a large vacuum box.

Data availability

The data that supports the findings of the study are included in the main text and supplementary information files or upon request from the corresponding authors. Source data are provided in this paper.

References

1. Xiong, Z. et al. Homogenized chlorine distribution for >27% power conversion efficiency in perovskite solar cells. *Science* **390**, 638–642 (2025).
2. Li, G. et al. Stabilizing high-efficiency perovskite solar cells via strategic interfacial contact engineering. *Nat. Photon.* **19**, 1-8 (2025).
3. Bi, L. et al. Temperature-controlled vacuum quenching for perovskite solar modules towards scalable production. *Nat. Photon.* **19**, 968–976 (2025).
4. Lian, X. et al. An inverted planar solar cell with 13% efficiency and a sensitive visible light detector based on orientation regulated 2D perovskites. *J. Mater. Chem. A.* **6**, 24633–24640 (2018).
5. Li, G. et al. Highly efficient pin perovskite solar cells that endure temperature variations. *Science* **379**, 399–403 (2023).
6. Zhang, H. et al. Lead immobilization for environmentally sustainable perovskite solar cells. *Nature* **617**, 687–695 (2023).
7. Li, Y. et al. Mutual stabilization of hybrid and inorganic perovskites for photovoltaics. *eScience* **5**, 100449 (2025).
8. Zhang, S. et al. Minimizing buried interfacial defects for efficient inverted perovskite solar cells. *Science* **380**, 404–409 (2023).
9. Yu, S. et al. Homogenized NiO_x nanoparticles for improved hole transport in inverted perovskite solar cells. *Science* **382**, 1399–1404 (2023).

10. Li, Z. et al. Stabilized hole-selective layer for high-performance inverted p-i-n perovskite solar cells. *Science* **382**, 284–289 (2023).
11. Ou, Y. et al. Collaborative interfacial modification and surficial passivation for high-efficiency MA-free wide-bandgap perovskite solar cells. *Chem. Eng. J.* **469**, 143860 (2023).
12. Al-Ashouri, A. et al. Conformal monolayer contacts with lossless interfaces for perovskite single junction and monolithic tandem solar cells. *Energy Environ. Sci.* **12**, 3356–3369 (2019).
13. Lan, Z. et al. Self-assembled monolayers as hole-transporting materials for inverted perovskite solar cells. *Mol. Syst. Des. Eng.* **8**, 1440–1455 (2023).
14. Zhao, K. et al. peri-Fused polyaromatic molecular contacts for perovskite solar cells. *Nature* **632**, 301–306 (2024).
15. Qu, G. et al. Self-assembled materials with an ordered hydrophilic bilayer for high performance inverted perovskite solar cells. *Nat. Commun.* **16**, 86 (2025).
16. Zhu, J. et al. Self-assembled hole-selective contact for efficient Sn-Pb perovskite solar cells and all-perovskite tandems. *Nat. Commun.* **16**, 240 (2025).
17. Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
18. Li, D. et al. Co-adsorbed self-assembled monolayer enables high-performance perovskite and organic solar cells. *Nat. Commun.* **15**, 7605 (2024).
19. Zhang, X. et al. Reinforcing coverage of self-assembled monomolecular layers for inverted perovskite solar cells with efficiency of 25.70 %. *Angew. Chem. Int. Ed.* **64**, e202423827 (2025).
20. Qu, G. et al. Conjugated linker-boosted self-assembled monolayer molecule for inverted perovskite solar cells. *Joule* **8**, 2123–2134 (2024).
21. Yu, B. et al. Minimizing buried interface energy losses with post-assembled chelating molecular bridges for high-performance and stable inverted perovskite solar cells. *Adv. Mater.* **37**, 2500708 (2025).
22. Ren, B. et al. A Universal p-type heterointerface for inverted perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202507950 (2025).
23. Ma, J. et al. Homogenizing self-assembled monolayers via covalent organic frameworks for inverted

perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e19875 (2025).

24. Shi, C. et al. Modulating competitive adsorption of hybrid self-assembled molecules for efficient wide-bandgap perovskite solar cells and tandems. *Nat. Commun.* **16**, 3029 (2025).
25. Wang, L. et al. In situ crosslinked self-assembled monolayer as processing stable hole selective contact in perovskite solar cells. *Angew. Chem. Int. Ed.* **147**, e17058 (2025).
26. Zhang, X. et al. A spiro-type self-assembled hole transporting monolayer for highly efficient and stable inverted perovskite solar cells and modules. *Energy Environ. Sci.* **18**, 468–477 (2025).
27. Truong, M. A. et al. Tripodal triazatruxene derivative as a face-on oriented hole-collecting monolayer for efficient and stable inverted perovskite solar cells. *J. Am. Chem. Soc.* **145**, 7528–7539 (2023).
28. Yuan, C. et al. Strengthening durability of electron selective layer and interface via trifluoromethoxy-functionalized bguanide cation toward high-performance air-processed n-i-p perovskite solar cells. *Adv. Mater.* **37**, e18388 (2025).
29. Liu, B. et al. Polydentate ligand reinforced chelating to stabilize buried interface toward high-performance perovskite solar cells. *Angew. Chem. Int. Ed.* **63**, e202317185 (2024).
30. Ding, J. et al. Stabilizing buried interface by bilateral bond strength equilibrium strategy toward efficient perovskite photovoltaics. *Nat. Commun.* **16**, 8407 (2025).
31. Zhang, H. et al. Multimodal host–guest complexation for efficient and stable perovskite photovoltaics. *Nat. Commun.* **12**, 3383 (2021).
32. Lian, X. et al. A supramolecular approach to improve the performance and operational stability of all-perovskite tandem solar cells. *Nat. Commun.* **16**, 7173 (2025).
33. Yang, M. et al. Reducing lead toxicity of perovskite solar cells with a built-in supramolecular complex. *Nat. Sustain.* **6**, 1455–1464 (2023).
34. Guo, H. et al. Immobilizing surface halide in perovskite solar cells via calix[4]pyrrole. *Adv. Mater.* **35**, 2301871 (2023).
35. He, D. et al. Universal ion migration suppression strategy based on supramolecular host–guest interaction for high-performance perovskite solar cells. *Adv. Mater.* **37**, 2505115 (2025).
36. Liu, X. et al. Reproducible fabrication of perovskite photovoltaics via supramolecule confinement growth. *Nano-Micro Lett.* **18**, 67 (2025).

37. Su, T.-S. et al. Supramolecular engineering in hybrid perovskite optoelectronics. *Chem. Soc. Rev.* **54**, 6448-6481 (2025).
38. Wang, Z. et al. Buried interface regulation with a supramolecular assembled template enables high-performance perovskite solar cells for minimizing the V_{oc} deficit. *Adv. Mater.* **37**, 2418011 (2025).
39. Fang, Z. et al. Grain boundary engineering with self-assembled porphyrin supramolecules for highly efficient large-area perovskite photovoltaics. *J. Am. Chem. Soc.* **143**, 18989-18996 (2021).
40. Song, H. et al. Supramolecular design principles in pseudohalides for high-performance perovskite solar mini modules. *Joule* **8**, 2283-2303 (2024).
41. Zhang, Z. et al. Coordination engineering with crown ethers for perovskite precursor stabilization and defect passivation. *Energy Environ. Sci.* **17**, 7182-7192 (2024).
42. Wang, J. et al. Synergistic iodine and lead chelation with redox cycling via supramolecular engineering for stable and sustainable perovskite solar cells. *Joule* **9** 102105 (2025).
43. Shen, Y. et al. Crowning lithium ions in hole-transport layer toward stable perovskite solar cells. *Adv. Mater.* **34**, 2200978 (2022).
44. Zhang, H. et al. Excess PbI_2 management via multimode supramolecular complex engineering enables high-performance perovskite solar cells. *Adv. Energy Mater.* **12**, 2201663 (2022).
45. Long, Z. et al. Supramolecular cucurbit[5]uril modulates the buried SnO_2 /perovskite interface for efficient and stable perovskite solar cells. *Adv. Funct. Mater.* **34**, 2408818 (2024).
46. Yang, G. et al. Tailored supramolecular interactions in host–guest complexation for efficient and stable perovskite solar cells and modules. *Angew. Chem. Int. Ed.* **63**, e202410454 (2024).
47. Chen, X. et al. Supramolecular Aza Crown Ether Modulator for Efficient and Stable Perovskite Solar Cells. *Adv. Funct. Mater.* **34**, 2311527 (2024).
48. Yang, Y. et al. Passivation of positively charged cationic defects in perovskite with nitrogen-donor crown ether enabling efficient perovskite solar cells. *Chem. Eng. J.* **451**, 138962 (2023).
49. Xu, G. et al. Amidinopyridine ion docking in crown ether cavity to modulate the top interface in inverted perovskite solar cells. *Adv. Energy Mater.* **15**, 2405088 (2025).
50. Zhao, C. et al. Stabilization of highly efficient perovskite solar cells with a tailored supramolecular interface. *Nat. Commun.* **15**, 7139 (2024).

51. Su, T.-S. et al. Crown ether modulation enables over 23% efficient formamidinium-based perovskite solar cells. *J. Am. Chem. Soc.* **142**, 19980-19991 (2020).
52. Sadhu, A. et al. Elucidating the role of chalcogenide-based interface passivators in enhancing the stability of perovskite solar cells. *Adv. Funct. Mater.* **33**, 2305215 (2023).
53. Wharf, I. et al. Synthesis and vibrational spectra of some lead (II) halide adducts with O-, S-, and N-donor atom ligands. *Can. J. Chem.* **54**, 3430-3438 (1976).
54. Yang, Y. et al. Passivation of positively charged cationic defects in perovskite with nitrogen-donor crown ether enabling efficient perovskite solar cells. *Chem. Eng. J.* **451**, 138962 (2023).

Fundings

This work was financially supported by the National Natural Science Foundation of China (52462031, 62274018), Xinjiang Construction Corps Key Areas of Science and Technology Research Project (2023AB029), Photovoltaic-Storage Integrated Application Functional Materials and Technology Science and Technology Innovation Team of Department of Education of Yunnan Province, Key Project of Chongqing Overseas Students Returning to China Entrepreneurship and Innovation Support Plan (cx2023006), Yunnan fundamental Research Projects (202501AU070122), and Kunming University of Science and Technology Analysis and Testing Fund (2024T20230245). This work was also supported by the National Natural Science Foundation of China (22279149), and Youth Innovation Promotion Association of the Chinese Academy of Sciences (2022034).

Author contributions

J.C., Y.Y. and R.Y. conceived the idea. R.Y. fabricated and characterized the perovskite films and devices. J.D., H.L. and C.C. participated in device fabrication and certification. R.Y., J.D., X.H. and X.L. performed the stability measurements of films and devices. T.H., L.Y., C.H. and Y.H. fabricated large-area modules. S.G. and X.C. conducted TRS measurements. J.Z. performed theoretical calculations. K.Z. and Y.W. assisted in the preparation of all-perovskite tandem solar cell. M.F. and X.G. participated in the implementation of GIXRD testing. Z.L., X.L., D.H., and X.S. were involved in the data analysis. R.Y. and

Y.Y. wrote the first draft of the manuscript. J.C. wrote and revised the final version of the manuscript. J.C. and J.Y. supervised the project. All authors contributed to the discussion and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Editorial summary:

Self-assembled interlayers can improve efficiency in inverted perovskite solar cells, but poor coverage and weak interfaces limit stability. Yang et al stabilized the buried interface using supramolecular host–guest chemistry with functionalized calixarene molecules, improving cell’s performance.

Peer review information: *Nature Communications* thanks Nam Jeon, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.